

Clean Schemas, Stereotyped Personas: Locating Cultural Bias in Persona-Conditioned Language Models

Author: Bianca Dené Williams **Affiliation:** Independent Researcher **ORCID:** 0009-0004-6378-628X **Date:** August 2026 **Version:** 7

Abstract

Persona-based evaluation systems simulate diverse users by conditioning language models on structured identity attributes. This practice is audited at two levels: the schema and dependency graph of MatrAIx, an open-source evaluation framework, and the behavior of personas rendered from it, across 1,175 trials on two model families, using two independent coding methods.

Four findings are reported. First, cultural background labels produce large, systematic behavioral shifts: personas labeled “Individualist (Western)” used direct assertion in 96% of trials versus 24% for “Collectivist (East Asian),” with permission-seeking language at 0% versus 60%. A blind LLM judge independently reproduced this pattern. Second, schema analysis found no mechanism for this effect within the persona system: across five identity dimensions there are zero edges to competence dimensions, and no cultural-background edge differentiates cultural values by more than 0.0111, with the four edges carrying documented cross-cultural rationale differentiating them not at all. The stereotyping originates in model priors at render time, not in persona system design.

Third, replication of the full cultural-background probe on a second provider (GPT-4o, n=200) found comparable separation across cultural values but no detectable rank agreement with the first model on any measured dimension. The phenomenon generalizes; the specific profile of stereotyping does not. An audit conducted on one model does not transfer to another.

Fourth, framing salience moderates the effect modestly. In a controlled comparison holding task, model, conditioning, and cue length constant, directing reflection toward cultural background rather than general workplace habit narrowed the directness and permission-seeking gaps under both coding methods, though estimates diverge substantially (lexical: 35% and 40%; blind judge: 14% and 12%), and the judge found two other dimensions widening.

Two non-replications are additionally reported. A pilot finding that a lower-resource language label suppressed apparent competence did not survive expansion from n=5 to n=25, and native-language labels showed no representation-ordered effect at n=175. My own interpretive claim that orientation-bearing cultural labels polarize most, advanced in an earlier version of this paper on single-model evidence, did not survive cross-model testing and is withdrawn.

The divergence between the two model families is a construct validity problem and not only a stability problem: two instruments that rank the same categories with no agreement are not tracking a common underlying construct. The behaviors most often attributed to culture in these systems, directness, deference, hedging, and permission-seeking, are treated in the relevant literature as mitigation strategies governed by power distance, social distance, and size of imposition within a specific interaction, all of which a persona system can specify directly. Cultural stereotyping in persona conditioning cannot be addressed through schema design alone, must be validated against each model a system actually deploys, and should not be presented as measurement of demographic groups without a demonstration of construct validity that the field has not yet required.

1. Introduction

Persona-based evaluation has become standard practice for testing AI products against simulated diverse users. Rather than recruiting participants across every demographic, developers generate structured persona records and instantiate them as language model agents. MatrAIx, an open-source framework, exemplifies the approach: a schema of 1,290 categorical dimensions, a dependency-aware generative graph of 6,999 directed edges, and a public release of one million curated personas.

When such systems include cultural or linguistic attributes, they make implicit claims about how identity shapes behavior. If a persona labeled “Collectivist (East Asian)” communicates differently from one labeled “Individualist (Western),” is the system reproducing documented cross-cultural variation, or amplifying a stereotype?

MatrAIx has infrastructure suited to answering this: a persona-adherence suite that runs positive/negative persona

pairs through neutral tasks across four environments and uses an LLM judge to assess whether the target attribute surfaces in behavior. That suite covers ten attributes, including coding style, communication traits, and register. It covers no identity attribute.

This paper approaches the gap from two directions. Behavior is measured across 1,175 counterfactual trials, and the schema and dependency graph are examined directly to determine whether observed effects are attributable to the persona system’s design. The two lines of evidence diverge, and that divergence is the central result of this paper.

A third line of evidence, cross-model replication, bears on how far that result travels. It travels in kind but not in detail: a second model family stereotypes just as systematically, and differently.

MatrAIX is the case study here rather than the subject. Because the effects reported here originate at generation time, they are inherited by any system that renders an identity label into a prompt, whatever the quality of the schema supplying it. That MatrAIX’s design turns out to be careful is what makes the dissociation visible; a less rigorous system would have confounded it. The recommendations are correspondingly split between those that apply to any system conditioning on identity and those specific to MatrAIX.

Two findings that did not replicate are also reported, one of them my own interpretive claim from an earlier version of this work. Both are described in full.

2. Methods

2.1 Design

Each probe varies a single identity attribute while holding all other persona content constant, then compares responses to an identical task. Task instructions never mention communication style, competence, or any measured dimension.

Persona conditioning uses MatrAIX’s own phrase templates verbatim, from `persona/schema/dimensions.json`: “with a {value} cultural frame” for `cultural_background`, “a native {value} speaker” for `primary_language`.

Probe 1, cultural background, implicit framing (n=200). All eight `cultural_background` schema values, 25 trials each. Personas expressed disagreement with a manager’s proposed project change. Cultural identity was supplied as a background attribute.

Probe 2, language label (n=175). Seven conditions, 25 trials each: six valid `primary_language` schema values spanning a range of training-corpus representation (English, Mandarin, Hindi, Arabic, Bengali, Swahili), plus Fulfulde as an out-of-schema control. Personas were software engineers with ten years of experience asked to evaluate a distributed system architecture.

Probe 3, cultural background, explicit framing (n=200). Same eight values, 25 trials each, with cultural identity foregrounded in the task. Personas challenged a senior colleague’s flawed proposal.

Probe 4, framing isolation (n=400). Probes 1 and 3 differed in both task and framing, confounding interpretation. Probe 4 holds the task fixed (the Probe 1 manager-disagreement scenario) and varies only the object of a reflection cue:

- Neutral cue: “Consider how you typically handle disagreement at work.” (9 words)
- Cultural cue: “Consider how your cultural background shapes the way you handle disagreement.” (11 words)

Both conditions receive persona conditioning, the same task, and a reflection instruction of comparable length. Only the object of reflection differs. Eight values by 2 conditions by 25 trials.

Probes 1 through 4 used Claude Opus 4.8 (`claude-opus-4-8`), default temperature, 300 max tokens. Subtotal: 975 trials.

2.2 Analysis

Coding was deterministic and lexical: marker categories defined a priori as regular expression sets, applied identically across conditions. Reported values are the percentage of trials containing at least one marker in a category. Coding scripts and full marker definitions are in the accompanying repository; the analysis involves no model judgment and is fully reproducible.

Blind LLM-judge coding was additionally applied (`claude-opus-4-8`) using a fixed six-dimension rubric on a 1 to 5 scale, to a subset of 10 trials per condition cell. The judge sees only response text and is never told the cultural background

or framing condition, preventing it from scoring the label rather than the text. Where the two methods diverge, both are reported.

Probe 4 additionally codes a `meta_commentary` category, language in which the response explicitly declines to caricature (“cultures vary widely,” “not a monolith,” “I won’t flatten this”), to test whether framing operates through explicit reasoning.

This approach is crude by design. It cannot capture meaning, irony, or implicature, and will miss markers phrased outside defined patterns. That is accepted in exchange for reproducibility and freedom from judge bias.

2.3 Schema and dependency graph analysis

The analysis covered MatrAIX’s published schema (`persona/schema/dimensions.json`) and generative dependency graph (`persona/synthesis/graph/full_dag.json`, 24.7 MB, 6,999 directed edges). For five identity dimensions, `cultural_background`, `primary_language`, `english_proficiency`, `multilingualism`, and `demo_ethnicity_broad`, all outgoing edges were enumerated, documentation metadata recorded, conditional probability tables extracted, and maximum probability spread measured across identity values.

Conditionals are stored as pairwise conditional matrices, with rows indexed by source values and columns by target values. For each edge the range across source values is taken for every target outcome, and the largest is reported, which is the most a single identity value can shift any one downstream probability. The script that reproduces this analysis from a local MatrAIX clone is `analyze_schema.py` in the accompanying repository.

2.4 Cross-model replication

To test whether Probe 1’s effects are specific to the model used, Probe 1 was repeated in full on a second provider: all eight `cultural_background` values, 25 trials each, on GPT-4o (`gpt-4o`), using the identical prompt, an identical 300-token limit, and default temperature. Total: 200 trials, all completed without error. Lexical coding used the same marker definitions applied to the Claude corpus, unmodified. Combined total across both models: 1,175 trials.

Replication on Gemini was also attempted (`gemini-3.5-flash`). API quota was exhausted after 17 trials from a single condition, which is not a usable comparison. The attempt is reported; no Gemini results are.

One methodological adjustment was required. GPT-4o produces typographic apostrophes (U+2019) in roughly a third of instances, while Claude produced none across 975 trials. Because several marker patterns contain a straight apostrophe, uncorrected coding systematically undercounted GPT-4o. Curly apostrophes were normalized to straight before coding. The correction shifted individual cell values by up to four percentage points and changed no reported direction. Claude results are unaffected, since coding was uniform within that corpus.

The two models are compared by per-value marker rates and Spearman rank correlations across the eight cultural values. With eight values, rank correlation has low power; it is treated as descriptive and significance is not interpreted.

3. Results

3.1 Probe 1: Cultural background under implicit framing (n=200)

Table 1. Percentage of trials containing at least one marker, Claude Opus 4.8 (n=25 per value).

Cultural background	Directness	Permission-seeking	Group reference	Hedging	Deference
Individualist (Western)	96%	0%	64%	96%	52%
Mixed / diaspora	64%	12%	20%	100%	12%
Latin	56%	28%	68%	100%	8%
Indigenous	44%	12%	48%	92%	12%
South Asian	44%	76%	40%	88%	28%
African	36%	72%	56%	96%	20%
Collectivist (East Asian)	24%	60%	96%	100%	48%
Middle Eastern	24%	64%	60%	100%	32%

The extremes on directness are the two values that name a psychological orientation: 96% versus 24%, a fourfold difference. Permission-seeking shows a categorical split: 0% for Individualist versus 60% to 76% for Collectivist, African, and South Asian. Group reference peaks at 96% for Collectivist versus 20% for Mixed / diaspora.

The orientation-free values (African, South Asian, Latin, Middle Eastern, Indigenous) occupy intermediate positions on directness while the two orientation-bearing labels hold the poles. Section 4.2 discusses the implication and reports that this pattern did not survive cross-model testing.

3.2 Probe 4: Framing salience, isolated (n=400)

Holding task, model, persona conditioning, and reflection-cue length constant, and varying only whether the cue directed attention to cultural background. Results from both coding methods are reported, which diverge in magnitude.

Table 2. Individualist versus Collectivist gap, lexical coding.

Marker	Neutral cue	Cultural cue	Change
Directness	92 pts	60 pts	-32 pts (-35%)
Permission-seeking	40 pts	24 pts	-16 pts (-40%)
Group reference	60 pts	56 pts	-4 pts (-7%)

Table 3. Individualist versus Collectivist gap, LLM-judge coding (1 to 5 scale, n=10 per cell, judge blind to both cultural background and framing condition).

Dimension	Neutral cue	Cultural cue	Change
Directness	2.80	2.40	-0.40 (-14%)
Permission-seeking	1.70	1.50	-0.20 (-12%)
Hedging	1.90	1.60	-0.30 (-16%)
Deference	1.90	2.10	+0.20 (+11%)
Group orientation	1.10	1.30	+0.20 (+18%)
Meta-awareness	0.50	0.10	-0.40 (-80%)

Convergence and divergence. Both methods find the directness and permission-seeking gaps narrowing under the cultural cue. That agreement, from an independent coding method whose judge never saw the condition labels, supports the direction of the effect.

The methods disagree substantially on magnitude: lexical coding shows reductions of 35% and 40%, judge coding 14% and 12%. The judge additionally finds the deference and group-orientation gaps widening by 11% and 18%, directions the lexical coding did not detect.

The two methods also disagree on meta-commentary. Lexical coding found explicit refusals to caricature rising under the cultural cue (4% to 16% for the Individualist condition). Judge coding found the gap in meta-awareness collapsing by 80% and overall spread narrowing by 28%, indicating more even distribution rather than a targeted rise. The reading of explicit reasoning as mechanism is therefore weakly supported at best.

Interpretation. Probes 1 and 3 suggested a framing effect but differed in task. Probe 4 removes that confound, and the direction of the effect on directness and permission-seeking survives independent coding. However, the effect is smaller than lexical coding alone indicated, is not uniform across dimensions, and moves in the opposite direction on two of six judged dimensions. The effect is directionally supported but modest, and is not claimed to constitute a reliable mitigation.

3.3 Probe 3: Cultural background under explicit framing (n=200)

Probe 3, though task-confounded relative to Probe 1, showed the same directional pattern: the Individualist/Collectivist directness gap fell from 72 to 36 points and permission-seeking from 60 to 32 points. It is reported as consistent supporting evidence rather than as independent confirmation, given the confound Probe 4 was designed to remove.

3.4 Probe 2: Language label (n=175), null result

Table 4. Percentage of trials containing at least one marker (n=25 per language).

Language				Directness
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	In schema	Task engagement	Competence hedging	
English	yes	100%	20%	52%
Mandarin	yes	100%	20%	40%
Hindi	yes	100%	36%	32%
Arabic	yes	100%	0%	64%
Bengali	yes	100%	12%	52%
Swahili	yes	100%	12%	64%
Fulfulde	no	100%	32%	68%

Task engagement was 100% across every condition. Competence hedging did not order by training-data representation: Hindi, a high-resource language, was highest at 36%, exceeding Fulfulde (32%). English showed 20%, above both Bengali and Swahili (12%). Arabic showed 0%.

Non-replication. A preliminary version at $n=5$ per condition found three of five Fulfulde-labeled responses opening with meta-cognitive disclosure and not engaging the technical task, a pattern absent from English and Mandarin. At $n=25$, Fulfulde task engagement was 100% and its competence hedging was exceeded by Hindi. The pilot result was not representative, and the representation-gradient hypothesis is not supported by these data.

3.5 Schema analysis: cultural background

cultural_background is dimension index 24, category “Demographic: Cultural,” described as “Cultural frame of reference,” rendered via “with a {value} cultural frame”. Its eight permitted values:

Individualist (Western), Collectivist (East Asian), South Asian, Latin, African, Middle Eastern, Indigenous, Mixed / diaspora

These operate at incommensurate conceptual levels. Two fuse a psychological orientation with a geographic region. Four are geographic or ethnocultural categories carrying no orientation. One denotes a political and colonial relationship. One denotes migration and mixed heritage.

The asymmetry constrains representable personas: a persona cannot be South Asian and individualist, or Western and collectivist, because orientation is bundled into two values and absent from the other six. The schema already models orientation independently elsewhere, via *decision_style*, *values_priority*, and *schwartz_value_conformity*.

3.6 Dependency graph analysis

The graph contains 6,999 directed edges. Well-documented edges carry written rationale, evidence level, relationship basis, edge weight, calibrated conditional probability table, and explicit epistemic hedging, for instance “direction_semantics”: “sampling direction; not necessarily an identified causal effect”.

Identity dimensions are documented unevenly.

primary_language: 18 edges, all documented. Twelve connect to corresponding language-proficiency dimensions (strength high, confidence 0.82), largely tautological. Six connect to communication-style dimensions: *register*, *cog_use_of_jargon*, *cog_precision_of_language*, *cog_formality*, *tone_expected*, *modality_pref* (strength medium, confidence 0.55).

cultural_background: 137 edges, 4 documented. The documented four connect to *values_priority*, *political_lean*, *att_traditional_gender_roles*, and *att_organized_religion* (strength medium, confidence 0.62), constructs with substantial cross-cultural literature. The remaining 133 contain only *edge_id*, *source*, *target*, *edge_weight*, and *cpd*: no rationale, no relation type, no evidence level, no strength, no confidence. Their targets are consumer attitudes and hobby interests, including *att_vaccines*, *att_gun_ownership*, *topic_woodworking*, *topic_chess*, and *topic_birdwatching*.

demo_ethnicity_broad: 0 outgoing edges.

Effect magnitude. For each of the 133 undocumented cultural-background edges, the maximum probability difference was measured between any two cultural values. The largest was 0.0111, approximately one percentage point. None exceeded 0.02. For *topic_woodworking*, seven of eight values produced numerically identical distributions. These edges are structurally present but functionally inert.

The four documented cultural-background edges are flatter still. All four, *values_priority*, *political_lean*, *att_traditional_gender_roles*, and *att_organized_religion*, produce numerically identical conditional distributions across all eight cultural values: a maximum spread of exactly zero. These are the edges carrying written rationale, medium strength, and confidence 0.62, connecting to constructs with substantial cross-cultural literature. They are

documented as though they encode cultural variation and are calibrated so that they encode none.

Taken together, no `cultural_background` edge anywhere in the graph differentiates one cultural value from another by more than 0.0111, and the four edges built with the most care differentiate them not at all.

Table 5. Identity dimension edges and competence connections.

Identity dimension	Total edges	Documented	Edges to competence dimensions
primary_language	18	18	0
english_proficiency	7	7	0
multilingualism	6	6	0
cultural_background	137	4	0
demo_ethnicity_broad	0	not applicable	0

No identity dimension connects to any `skill_*`, `fam_*`, `prog_*`, or `tool_*` dimension, nor to `highest_education`, `tech_savviness`, `institution_tier`, `academic_field`, `seniority`, or `research_output`.

3.7 Cross-model replication (n=200)

Table 6. Percentage of trials containing at least one marker, GPT-4o (n=25 per value).

Cultural background	Directness	Permission-seeking	Group reference	Hedging	Deference
African	36%	4%	100%	100%	4%
Individualist (Western)	36%	12%	48%	100%	0%
Mixed / diaspora	36%	4%	68%	100%	0%
Indigenous	32%	0%	68%	100%	8%
Collectivist (East Asian)	20%	16%	100%	100%	0%
Latin	12%	8%	48%	52%	4%
Middle Eastern	8%	8%	88%	100%	24%
South Asian	8%	8%	84%	100%	12%

The effect is present on both models. Cultural labels produced substantial separation across values on GPT-4o as they did on Claude: 28 points on directness, 52 on group reference, 48 on hedging. As with Claude, no schema mechanism is available to produce this. The label is rendered into the prompt and the model does the rest.

The profile of the effect does not transfer.

Table 7. Spread across the eight cultural values, and rank agreement between models.

Dimension	Claude spread	GPT-4o spread	Spearman rho	p
Directness	72 pts	28 pts	+0.48	0.22
Permission-seeking	76 pts	16 pts	+0.03	0.94
Group reference	76 pts	52 pts	+0.04	0.93
Hedging	12 pts	48 pts	-0.35	0.39
Deference	44 pts	24 pts	-0.16	0.70

There is no detectable rank agreement between the two models on any dimension. Directness shows the strongest association at $\rho = +0.48$, which at $n=8$ is not distinguishable from chance. Every other dimension is at or near zero.

Concretely: on Claude, South Asian was the second-highest value for permission-seeking at 76%; on GPT-4o it is 8%, tied near the bottom. On Claude, Latin ranked third on directness at 56%; on GPT-4o it is second from last at 12%. Middle Eastern is low on directness in both, and Individualist is high in both, but these are isolated agreements within an otherwise uncorrelated ordering.

Two coding notes. Competence hedging returned 0% across all 200 GPT-4o trials and remains uninterpretable; task engagement markers are Probe 2's technical vocabulary and are not expected to fire here. Deference returned 0% for Individualist (Western) and Collectivist (East Asian) but fired on four other values, peaking at 24% for Middle Eastern, so the category functions on this model.

One anomaly. Latin returned 52% hedging where the other seven values all returned 100%. Nothing in the Claude data anticipates this, and no account of it is available here. It may reflect a marker-fit artifact specific to how GPT-4o renders that label, or a genuine behavioral difference. It is reported as unexplained.

Interpretation. The paper’s central claim is that cultural stereotyping in persona conditioning originates in model priors at render time rather than in persona system design. Replication on a second provider, with different training data, an identical prompt, and no schema involvement, supports that claim more strongly than a single-model result could.

The replication also shows that the specific profile of the effect is a property of the model, not of the cultural labels. The rates reported in Section 3.1 should be read as facts about Claude Opus 4.8. What generalizes is that conditioning on a cultural label produces systematic behavioral separation. What does not generalize is which label produces which behavior, or how large the separation is on any given dimension.

This has a practical consequence. A persona system audited on one model cannot assume its findings hold on another. The stereotyping is inherited from whichever model renders the persona, and different models stereotype differently.

4. Discussion

4.1 The behavioral effect is real; the schema does not produce it

Probe 1 documents a fourfold difference in direct assertion and a categorical difference in permission-seeking, driven solely by a cultural label rendered into a prompt. The dependency graph offers no mechanism. The 133 undocumented cultural-background edges connect to hobbies and consumer attitudes, not communication style, and their conditionals are near-uniform. The four documented edges are exactly uniform.

The available claim is therefore stronger than a comparison of magnitudes. It is not that the schema’s cultural effects are too small to account for a seventy-two-point behavioral gap. It is that the schema encodes no cultural differentiation at all: across 137 edges, the largest difference any cultural value makes to any downstream probability is 0.0111, and the edges built with documented cross-cultural rationale make no difference whatsoever. There is no schema contribution to subtract. The entire observed effect arises at render time.

The effect originates downstream of the schema, in the model’s interpretation of an identity label at generation time. Any persona system rendering a cultural label into a prompt inherits this behavior regardless of schema quality. Section 3.7 shows this holds for a second model family whose training data and provider differ, which is what the claim requires.

4.2 Orientation-bearing labels polarize most on Claude, but not on GPT-4o

On Claude, the two schema values naming a psychological orientation occupied the extremes on nearly every measured dimension, while the six geographic or relational values clustered between them. In an earlier version of this paper I interpreted this as the model responding to the orientation term, “Individualist” or “Collectivist,” rather than to cultural knowledge about the associated regions, and suggested that the construct conflation described in Section 3.5 supplies the specific lexical trigger for stereotyped output.

The GPT-4o replication does not support this. Individualist (Western) ties for highest directness at 36%, but with African and Mixed / diaspora, neither of which names an orientation. Collectivist (East Asian) sits mid-range at 20%, with Latin, Middle Eastern, and South Asian below it. The extremes on directness are South Asian and Middle Eastern at 8%. On group reference, Collectivist reaches 100%, but African matches it exactly and South Asian and Middle Eastern are close behind at 84% and 88%. The orientation-bearing labels are not distinctive on either dimension.

I therefore withdraw the causal reading. The polarization pattern is a property of Claude’s response to these particular strings, not a general consequence of naming an orientation in a cultural label.

The schema observation in Section 3.5 is unaffected, because it does not depend on behavioral data. The eight values still operate at incommensurate conceptual levels, and the schema still cannot represent a persona who is both South Asian and individualist, because orientation is bundled into two values and absent from six. That is a representational constraint visible on inspection. What can no longer be claimed is that resolving it would reduce stereotyping.

4.3 Framing salience moderates the effect, modestly

Probe 4 isolates what Probes 1 and 3 could only suggest. With task, model, conditioning, and cue length held constant,

directing reflection toward cultural background rather than general workplace habit narrowed the directness and permission-seeking gaps under both coding methods.

The convergence is directional, not quantitative. Lexical coding indicated reductions of 35% and 40%; blind judge coding indicated 14% and 12%. The smaller figure is the more conservative estimate, since the judge assesses whole responses rather than counting phrase matches and is less sensitive to which specific wordings the regex patterns happen to capture.

The effect is also not uniform. Judge coding found deference and group-orientation gaps widening under the cultural cue while directness and hedging narrowed. This complicates any simple account of framing reducing stereotyping. A more accurate description is that directing attention to cultural background redistributes which dimensions carry the cultural signal, with net narrowing on the two dimensions where polarization was largest.

I initially interpreted a rise in explicit meta-commentary, statements that cultures vary and should not be flattened, as evidence for a reasoning-based mechanism. Judge coding did not support this: meta-awareness became more evenly distributed rather than more frequent. The mechanism remains unestablished.

The practical implication is correspondingly narrower than I first supposed. Conditioning templates measurably affect stereotype magnitude and are therefore worth testing, but the intervention tested here is not large enough to be recommended as a mitigation on its own evidence. It is also single-model. Given that the Probe 1 profile did not transfer across models, the framing result should not be assumed to either.

4.4 On the two non-replications

My pilot finding, competence suppression for a lower-resource language label, was striking, thematically coherent, and wrong. Three of five responses showed the pattern; twenty-five trials showed no representation-ordered effect, and a high-resource language exceeded the lower-resource one.

My orientation-bearing-labels claim followed the same shape at a different scale. It was based on 200 trials rather than 5, was internally consistent, and connected neatly to a schema-level observation already made independently. It did not survive contact with a second model.

I report both at length because the failure mode seems likely to be common in persona research, and because the two cases fail differently. The first is a sample-size failure: five trials produced a pattern that did not exist. The second is a generalization failure: two hundred trials produced a pattern that does exist, on that model, and does not describe cultural labels as such. Adequate n protects against the first and not at all against the second. At each stage, the finding I found most compelling was the one that dissolved on expansion.

4.5 The cross-model result is a validity problem, not only an instability problem

Section 3.7 reports that two models produce comparable separation across the same eight cultural labels with no detectable rank agreement on any dimension. This has so far been treated as a practical warning: audit findings do not transfer between models. It supports a stronger reading.

In measurement terms, two instruments that rank the same eight categories with correlations at or near zero on every dimension are not measuring the same construct. If cultural background were a stable property that both models tracked with noise, the orderings would agree imperfectly rather than not at all. What is observed instead is that each model produces an ordering that is systematic within itself, reproducible across 25 trials per cell, and unrelated to the other model's.

The most economical account is that neither model is tracking cultural variation. Each is producing label-conditioned output governed by whatever associations that particular string carries in that particular model's priors. The output is stable enough to look like measurement and arbitrary enough, across instruments, to fail as measurement.

This matters because persona systems are used as measurement instruments. The purpose of generating a persona labeled "South Asian" is to learn something about how a product performs for South Asian users. That inference requires the label to track something real about the population it names. The data here provide no evidence that it does, and the cross-model divergence is evidence against.

The claim needs stating carefully. Two models were tested, on one probe, with one task, coded by one method whose portability has already been qualified. This does not establish that no persona system can achieve construct validity on cultural attributes. What it establishes is that construct validity has not been demonstrated for the practice as it currently operates, that the field has not asked for such a demonstration, and that the one test available here came

back negative.

4.6 The measured behaviors are governed by situational face-work, not group membership

There is a further reason to expect cultural labels to fail as predictors of these particular behaviors, visible in the design of Probe 1 itself.

The task asks a persona to disagree with a manager's proposed project change. That is a face-threatening act performed under power asymmetry. The dimensions coded here, directness, deference, hedging, and permission-seeking, are exactly the linguistic resources that politeness research treats as mitigation strategies selected according to features of the interaction: the power differential between speakers, their social distance, and the size of the imposition being made (Brown & Levinson, 1987). In that account the strategies themselves are general, and what varies is how the situational variables are weighted and what counts as a large imposition.

A schema that varies these behaviors by cultural background has therefore assigned to stable group membership what the relevant literature assigns to relational context. The same person disagrees differently with a manager, a peer, and a direct report, and differently again when the stakes are high than when they are low. None of that variance is captured by a cultural label, and all of it is captured by variables a persona system could specify directly.

This also predicts the pattern in Section 3.7. If the behaviors in question are governed by features of the interaction rather than by cultural membership, then there is no stable cultural signal for a model to track, and two models given the same labels have nothing to agree about. The uncorrelated orderings are what the absence of an underlying construct looks like from the outside.

The constructive implication is that the situational variables are available and testable in a way the cultural label is not. Power distance within the specific relationship, familiarity between the parties, the magnitude of the imposition, and the institutional register of the setting can each be varied counterfactually, and predictions about their effects on mitigation behavior can be checked against a substantial body of prior work. Section 6 develops this as a recommendation.

This argument has a limit. The situational variables were not manipulated here, so it cannot be shown that they predict the behavior better than the cultural label does. That comparison is the obvious next experiment and it was not run. What can be said is that the schema currently varies these behaviors along a dimension for which no cross-model construct validity was found, while holding constant the dimensions the literature treats as governing them.

5. Limitations

Coding methods disagree on magnitude. Deterministic lexical coding and blind LLM-judge coding agree on the direction of the Probe 1 cultural effect and on the direction of the Probe 4 framing effect, but diverge by roughly a factor of three on the size of the latter, and disagree on the sign for two dimensions. Neither method is validated against human coding with inter-rater reliability, which remains the appropriate next step. Where the methods diverge, both are reported and the more conservative estimate treated as primary.

Judge coding used a subset. Judge scores are based on 10 trials per cell rather than the full 25, for cost reasons. Lexical coding covers all trials.

Partial cross-model coverage. Probe 1 was replicated in full on GPT-4o (n=200, all eight values). Probes 2, 3, and 4, including the framing manipulation, remain single-model results on Claude Opus 4.8. A Gemini replication was attempted and abandoned when API quota was exhausted after a partial single-condition sample. The framing effect reported in Section 3.2 is therefore untested outside one model, and given that the Probe 1 profile did not transfer across models, it should not be assumed to.

Lexical markers are not portable across models without validation. The marker sets were developed against Claude output and applied uniformly within that corpus, which is what internal comparison requires. Applied to GPT-4o, most categories functioned, but competence hedging returned 0% across all 200 trials through pattern mismatch rather than behavioral absence, and one value returned an unexplained hedging rate. Cross-model lexical comparison requires per-model marker validation or a coding method less sensitive to surface phrasing; the blind judge rubric used in Probe 4 would likely transfer better and is the appropriate instrument for a fuller cross-model study. Orthographic differences between models, apostrophe encoding in this case, can also silently bias regex-based coding and should be normalized before any cross-model comparison.

Rank comparison is underpowered. Spearman correlations across eight cultural values cannot distinguish weak association from none. They are reported descriptively. The absence of detectable rank agreement is consistent with the per-value rates, which differ substantially, but is not itself a significance test.

English only. All interactions were in English, including for personas specified as native speakers of other languages.

No statistical testing. Only descriptive rates are reported. With 25 trials per cell, the Probe 1 and Probe 4 differences are unlikely to be noise, but no significance testing was conducted and no correction made for multiple comparisons.

Framing mechanism untested. Probe 4 establishes that cultural-directed reflection reduces polarization. It does not establish why. The meta-commentary result is suggestive but low-frequency.

Partial schema coverage. Five identity dimensions and the directed edge set. `undirected_factors`, `high_order_factors`, `latent_modules`, and `conditional_masks` were not analyzed.

Out-of-schema condition. The Fulfulde condition used a value outside `primary_language`'s permitted set, making it a test of language-label conditioning generally rather than of MatrAIx persona generation specifically.

The situational-variable alternative is untested. Sections 4.6 and 6.1 argue that the behaviors measured here are governed by features of the interaction rather than by cultural membership, and recommend specifying those features instead. Power distance, social distance, and imposition size were not manipulated, so it cannot be shown that they predict the behavior better than a cultural label does. That comparison is the obvious next experiment. The data support only the negative half of the argument: the cultural label showed no cross-model construct validity on these dimensions.

Construct validity is assessed on a narrow base. The validity argument in Section 4.5 rests on rank agreement between two models, on one probe, with one task, under one coding method whose cross-model portability has been separately qualified. It establishes that construct validity has not been demonstrated, not that it is unattainable.

6. Recommendations

MatrAIx is the case study here, not the subject. The effects reported here originate at render time and are inherited by any system that conditions a language model on an identity label, whatever the quality of the schema supplying it. What applies generally is therefore separated from what applies to MatrAIx specifically.

6.1 For any system conditioning a model on identity attributes

This includes persona-based evaluation frameworks, synthetic user research, simulated-participant UX testing, and agent systems with demographic character specifications. The relevant question is not what kind of system it is but whether an identity label is rendered into a prompt.

Validate rendered behavior, not schema structure alone. Schema review would not have detected any effect reported here. The strongest behavioral difference measured has no representation in the dependency graph, and the graph encodes no cultural differentiation at all.

Validate against each model actually deployed. Section 3.7 reports comparable stereotyping on a second provider with no rank agreement on any dimension. Identity-attribute validation results are model-specific and should not be assumed to survive a change of provider or, presumably, a change of model version.

Do not treat cultural-label output as measurement without demonstrating construct validity. The purpose of generating a persona labeled with a cultural category is normally to learn something about how a system performs for the population that category names. That inference requires the label to track something real about that population. The cross-model result here provides evidence against it, and no published work establishing it is known to me. Systems that present persona output as evidence about demographic groups should either demonstrate that validity or state that they have not.

Model the situational variables that govern the behavior, not the group label. The communication behaviors most often attributed to culture, directness, deference, hedging, and permission-seeking, are treated in the relevant literature as mitigation strategies selected according to power distance, social distance, and size of imposition within a specific interaction (Brown & Levinson, 1987). Those variables can be specified in a persona, varied counterfactually, and checked against prior work. A cultural label bundles them into a group category, discards the relational information that actually governs them, and, on this evidence, produces output that two models do not

agree about. Where a system needs to represent variation in communication style, specifying the interaction is both more tractable and more defensible than specifying the culture.

Treat the conditioning template as a design parameter. Probe 4 indicates that how identity enters the prompt affects stereotype magnitude independent of which attributes are included. Systems should test their own conditioning templates rather than assuming neutrality. The intervention tested here is too modest to recommend as mitigation. The finding is that the template is a surface worth testing, not that the right template has been found.

Report sample sizes prominently, and replicate across models. The pilot reported here demonstrates how readily small-n persona studies produce compelling non-findings. My own withdrawn interpretive claim demonstrates that adequate n does not protect against a pattern that is real but model-specific.

6.2 For MatrAIx specifically

Extend the persona-adherence suite to identity attributes. The infrastructure is suited to this and currently covers ten behavioral attributes and no identity attributes. `cultural_background` is rendered directly into prompts and demonstrably shifts behavior.

Separate cultural affiliation from cultural orientation. Orientation is already modeled independently in the schema, via `decision_style`, `values_priority`, and `schwartz_value_conformity`. Bundling it into two of eight cultural values prevents the schema from representing combinations real populations exhibit: a persona cannot be South Asian and individualist, or Western and collectivist. This is a representational limitation independent of any behavioral effect. My earlier suggestion that separating these constructs might also reduce stereotyping is not supported by the cross-model data reported in Section 4.2, and is not advanced here.

Document or remove the 133 bare edges, and confirm the intent of the four documented ones. The bare edges are inert today, but they are undocumented surface area in a graph that documents comparable edges thoroughly. Separately, the four documented `cultural_background` edges carry rationale, evidence level, and confidence 0.62 while producing identical conditional distributions across all eight cultural values. If that uniformity is deliberate, a note recording the decision would prevent the edges being read as encoding variation they do not encode. If it is a calibration artifact, it is worth catching.

7. Conclusion

Across 1,175 trials, two model families, and two independent coding methods, cultural background labels produce large, systematic shifts in persona communication behavior. There is no corresponding effect for language labels, and I report the non-replication of my own preliminary finding on that point.

None of these effects are traceable to the persona system's schema or dependency graph, which contains no identity-to-competence edges and no cultural-background edge that differentiates cultural values by more than 0.0111. The four edges built with documented cross-cultural rationale differentiate them by exactly zero. This dissociation is the central result: a persona system's design can be careful, documented, and free of encoded identity-competence assumptions while the personas it renders still stereotype substantially. The stereotyping originates in model priors at render time and is inherited by any system that conditions a model on cultural identity.

Cross-model replication sharpens this. A second provider produced comparable separation across the same eight cultural labels with no detectable rank agreement on any dimension. The phenomenon is general; the profile is not. Because the profile differs between models, an audit conducted on one model does not transfer to another, and the specific rates reported here should be read as properties of the models that produced them rather than as facts about the labels.

That divergence carries a further implication. Two instruments that order the same eight categories with correlations at or near zero on every dimension are not measuring a shared construct. Each model produces output that is systematic within itself and unrelated to the other's, which is what the absence of an underlying construct looks like from outside. Persona systems are nonetheless used as measurement instruments, on the assumption that a cultural label tracks something real about the population it names. This work found no evidence for that assumption and some against it.

There is a reason to expect this. The behaviors measured here, directness, deference, hedging, and permission-seeking, are the mitigation strategies that politeness research treats as selected according to power distance, social distance, and size of imposition within a particular interaction. A schema that varies them by cultural membership has

assigned to a stable group category what the literature assigns to relational context, and has discarded the variables that actually govern the behavior. Those variables can be specified in a persona and varied counterfactually. The cultural label cannot be validated in the same way, and on this evidence does not survive the attempt.

Framing salience moderates the effect, but modestly, unevenly, and on one model only. Directing the model to reason about cultural background rather than supplying it inertly narrowed the two largest polarization gaps under both coding methods, while widening two others under judge coding. The conditioning template is a design surface worth testing, but the specific intervention examined here does not, on this evidence, constitute a mitigation.

The primary implication stands regardless: persona systems must validate what their personas do, not only how their schemas are structured, and must do so against the models they actually deploy. Schema review would not have detected any effect reported here, and single-model validation would have produced a picture that a second model does not share. Where a system needs to represent variation in communication behavior, the interaction is the tractable thing to specify. The culture is not.

Code and Data Availability

All probe implementations, raw trial outputs (1,175 responses across two providers), lexical coding scripts and marker definitions, the schema and dependency graph analysis script (`analyze_schema.py`) with its output, cross-model comparison scripts, and per-condition results are available at:

<https://github.com/biancadene/persona-cultural-validity>

References

- Blodgett, S. L., Barocas, S., Daumé III, H., & Wallach, H. (2020). Language (technology) is power: A critical survey of “bias” in NLP. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 5454–5476.
- Bolukbasi, T., Chang, K.-W., Zou, J., Saligrama, V., & Kalai, A. (2016). Man is to computer programmer as woman is to homemaker? Debiasing word embeddings. *Advances in Neural Information Processing Systems*, 29, 4349–4357.
- Brown, P., & Levinson, S. C. (1987). *Politeness: Some Universals in Language Usage*. Cambridge University Press.
- Buolamwini, J., & Gebru, T. (2018). Gender shades: Intersectional accuracy disparities in commercial gender classification. *Proceedings of Machine Learning Research*, 81, 1–15.
- Chang, J., Li, X., Hao, Y., Hou, J., Huang, J., Wen, Q., Huang, S., Liu, Y., Liu, X., Fan, Y., & Wang, Y. (2026). MatrAIx: Simulating the world with 8.3 billion persona agents. *arXiv preprint arXiv:2608.04205*.
- Council of Europe. (2020). *Common European Framework of Reference for Languages: Learning, Teaching, Assessment. Companion Volume*. Council of Europe Publishing.
- Cummins, J. (2001). *Negotiating Identities: Education for Empowerment in a Diverse Society* (2nd ed.). California Association for Bilingual Education.
- Hall, E. T. (1989). *Beyond Culture*. Anchor Books.
- Hofstede, G. (2010). *Cultures and Organizations: Software of the Mind* (3rd ed.). McGraw-Hill.
- Krashen, S. D. (1982). *Principles and Practice in Second Language Acquisition*. Pergamon Press.
- Nisbett, R. E. (2003). *The Geography of Thought: How Asians and Westerners Think Differently and Why*. Free Press.
- Sap, M., Gabriel, S., Qin, L., Jurafsky, D., Smith, N. A., & Choi, Y. (2020). Social bias frames: Reasoning about social and power implications of language. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 5477–5490.
- Tannen, D. (1990). *You Just Don't Understand: Women and Men in Conversation*. William Morrow.